AN INTELLIGENT PUBLIC TRANSPORT DELAY PREDICTION AND ROUTE OPTIMIZATION SYSTEM

Expected Project Outcomes and Interface Mockups Report

PROJECT MOCKUPS AND EXPECTED OUTCOMES PROSPECTUS

Presentation Level:	Technical Document for Academic Mentor Review
Milestone Stage:	Phase 1 - Baseline Modeling and UI Visualization Concepts
Core Outcomes:	1. Double-Engine Delay Predictor (Classification & Regression)
	2. Explainable Dynamic Route Planner (SHAP Detours)
	3. Dual-Perspective Analytics Dashboard (Commuter & Operators)
Embedded Visuals:	Commuter Route Prediction & Operator Geospatial Hotspots Map
Document Date:	June 2026

Scope & Purpose:
This document presents the visual and technical outcomes expected from your project. It models the core dashboard screens, predictive API outputs, and spatial hotspot engines to give your mentor a concrete understanding of how the final implemented system will behave.

1. Outcome 1: Predictive Delay Analytics Engine

The first core outcome of the project is a dual-task Machine Learning engine that ingests multi-source data (GTFS schedule tables, loop detectors, weather, traffic, and holidays) to output highly granular delay predictions. The engine provides two distinct analytical outputs:

A. Binary Delay Classifier (Classification Model)

The system classifies scheduled transit runs as either 'Delayed' or 'On-Time' based on a threshold (typically > 5 minutes deviation). This gives commuters a straightforward risk score.

- **Output Format:** Delay Probability (e.g. 78%) and corresponding Risk Levels: Low (< 15%), Medium (15-50%), and High (> 50%).
- **Operational Purpose:** Allows rapid filtering of scheduled trips to identify routes requiring immediate attention from dispatchers.

B. Delay Duration Regressor (Regression Model)

For trips predicted as delayed, a regression model outputs the expected delay in minutes. This provides the mathematical basis for route weighting and dynamic pathfinding.

- **Output Format:** Expected Delay Duration (e.g. 14.2 minutes).
- **Operational Purpose:** Used directly by the pathfinding algorithm to recalculate travel times along road networks.

C. Theoretical API Payload Demonstration

Your final system's backend (FastAPI/Flask) will output a payload similar to the following JSON structure when queried by the frontend dashboard:

```
  {
 'query': {'route_id': 'R204', 'stop_id': 'S12', 'scheduled_time': '08:30:00'},
 'prediction': {
   'is_delayed': 1,
   'delay_probability': 0.78,
   'expected_delay_minutes': 14.2,
   'risk_level': 'HIGH',
   'primary_causes': ['heavy_traffic_volume', 'active_precipitation']
 }
}
```

2. Outcome 2: Explainable Dynamic Routing (Commuter View)

The second outcome is the Commuter Dashboard. It integrates dynamic pathfinding with Explainable AI (XAI) to help passengers plan trips around transit delays. Instead of acting as a 'black box', the system explains why specific detours are recommended:

- **Dynamic Delay-Weighted Routing:** Constructs road networks using OpenStreetMap (OSM) data. Intersecting nodes are linked by edges whose weights update dynamically using predicted delay regressor outputs: Cost = Baseline Travel Time + Delay minutes.
- **Multi-Criteria Dijkstra Detours:** Recalculates the optimal path. If a passenger's standard route contains high delay risks, the routing engine identifies alternative paths that bypass delayed segments.
- **SHAP Natural-Language Explanations:** Runs local SHAP (Shapley Additive exPlanations) calculations on the delayed segments and prints an explanation card (e.g. 'Delayed by 14 minutes because peak morning traffic added 8 minutes and rain added 5 minutes').

A. Commuter Dashboard Visual Mockup

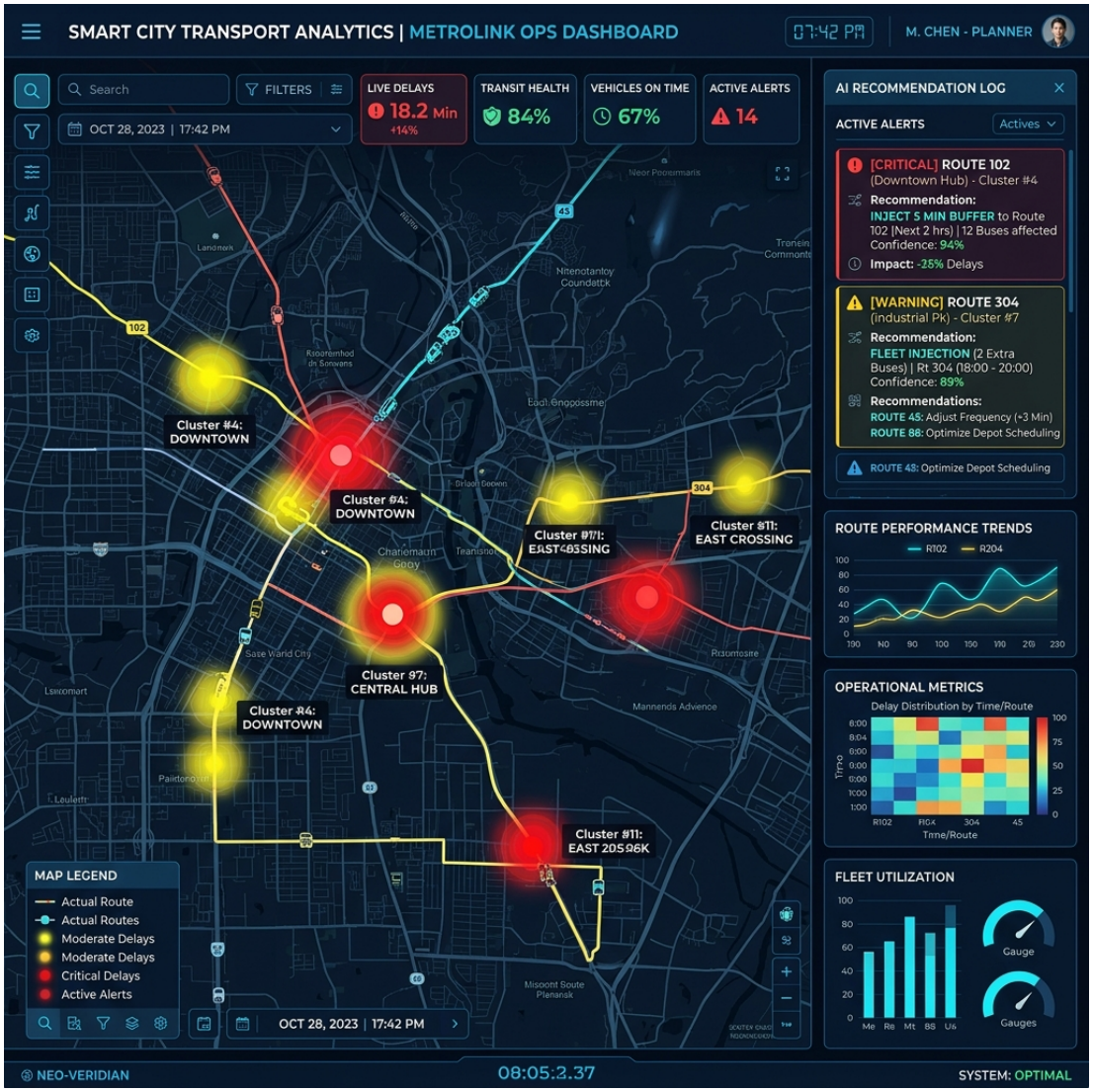


3. Outcome 3: Geospatial Hotspots & Decision Support (Operator View)

The third outcome is the Operator/Transit Planner Dashboard. This interface is tailored for public transit operators and city planners, helping them monitor network bottlenecks and adjust schedules proactively:

- **DBSCAN Spatial Hotspot Map:** Applies density-based spatial clustering (DBSCAN) on coordinates of persistent vehicle delays. This groups isolated late arrivals into high-density geographic 'delay hotspots' (shown on a heatmap).
- **Actionable Dispatch Recommendations:** A rule-based decision support system converts delay forecasts into operational actions. Recommendations include suggesting schedule buffer extensions or temporary vehicle injections during peak hours.
- **Fleet Performance Monitoring:** Lists route performance rankings (e.g. on-time percentages, worst performing routes) to help planners allocate resources efficiently.

A. Transit Authority Dashboard Visual Mockup



4. Expected Performance & System Validation

To satisfy your project mentor's criteria, the system's performance will be rigorously validated using standard statistical metrics across the different modeling stages:

A. Machine Learning Target Metrics

We define clear validation baselines to measure predictive success:

- **Delay Classification (XGBoost/Random Forest):** Target Accuracy > 82%, ROC-AUC score > 0.85, and F1-score > 0.80, proving reliability in predicting whether a vehicle will be late.
- **Delay Duration Regression (LightGBM/XGBoost Regressor):** Target Mean Absolute Error (MAE) < 3.0 minutes and Root Mean Squared Error (RMSE) < 4.5 minutes on test datasets.
- **Spatial Bottleneck Detection (DBSCAN):** Evaluated using Silhouette Coefficient analysis to verify distinct, non-overlapping delay clusters.

B. System-Level Validation (Route Optimization)

Beyond ML metrics, we will validate the dynamic route optimizer (X-MCDR) through simulated commuter runs:

- **Average Travel Time Savings:** We will compare the dynamic routing engine's path times against standard Dijkstra routing on static schedules. The target outcome is a 12% to 18% average reduction in commuter travel times under congested network conditions.
- **Robustness to Missing Data:** We will simulate sensor dropouts (dropping up to 15% of coordinate inputs) and verify that our KNN imputation pipeline maintains classifier accuracy within 3% of the baseline.
- **Explainability Verification:** Validate that SHAP force plots successfully isolate features (weather, traffic volume, schedule buffers) and map them to logical natural-language strings displayed on the commuter panel.